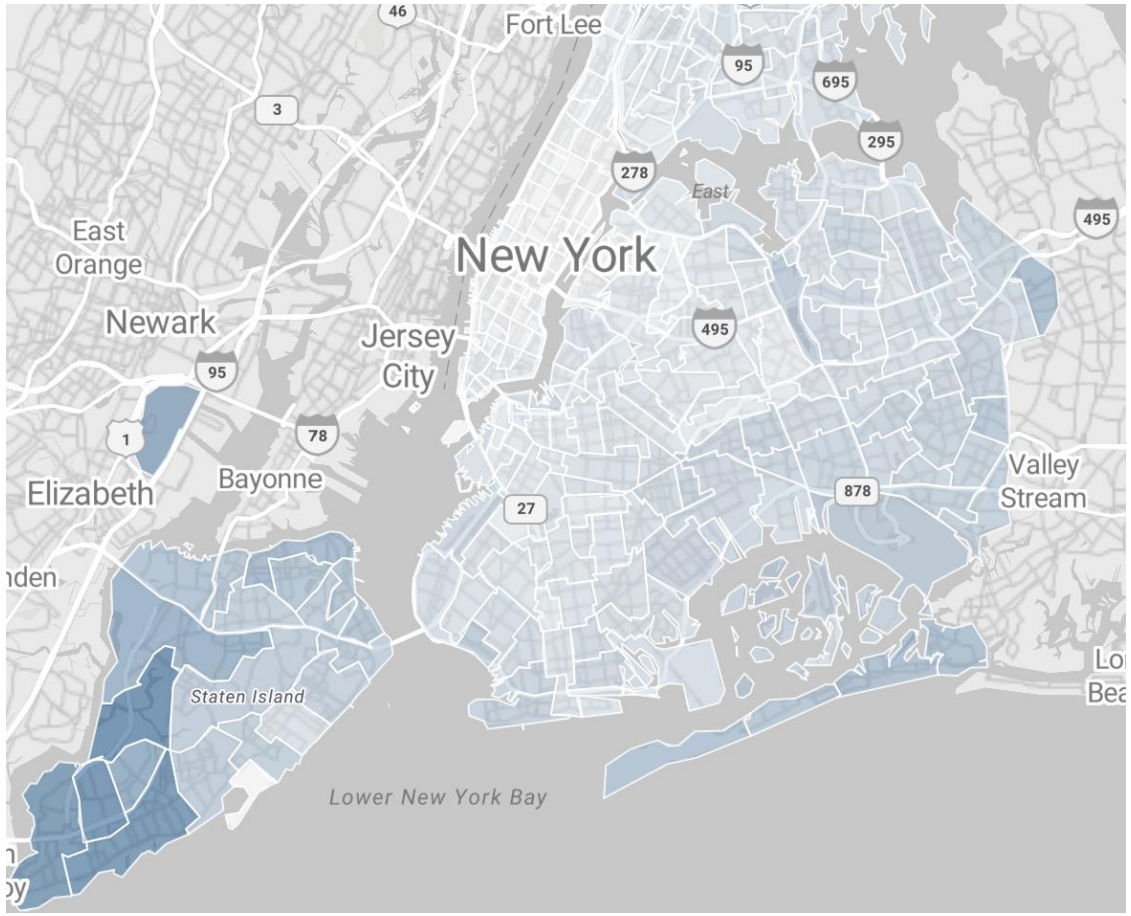
NYC Yellow Taxi: when, where and how drivers earn

Power BI analysis of 891,523 trips

May–June 2020 · Angel Iana



The data: 891,523 trips, checked in SQL and matched to TLC counts

898,131 trips
TLC trip records, 2 monthly files

→

SQL Server checks
6,608 trips with impossible values excluded

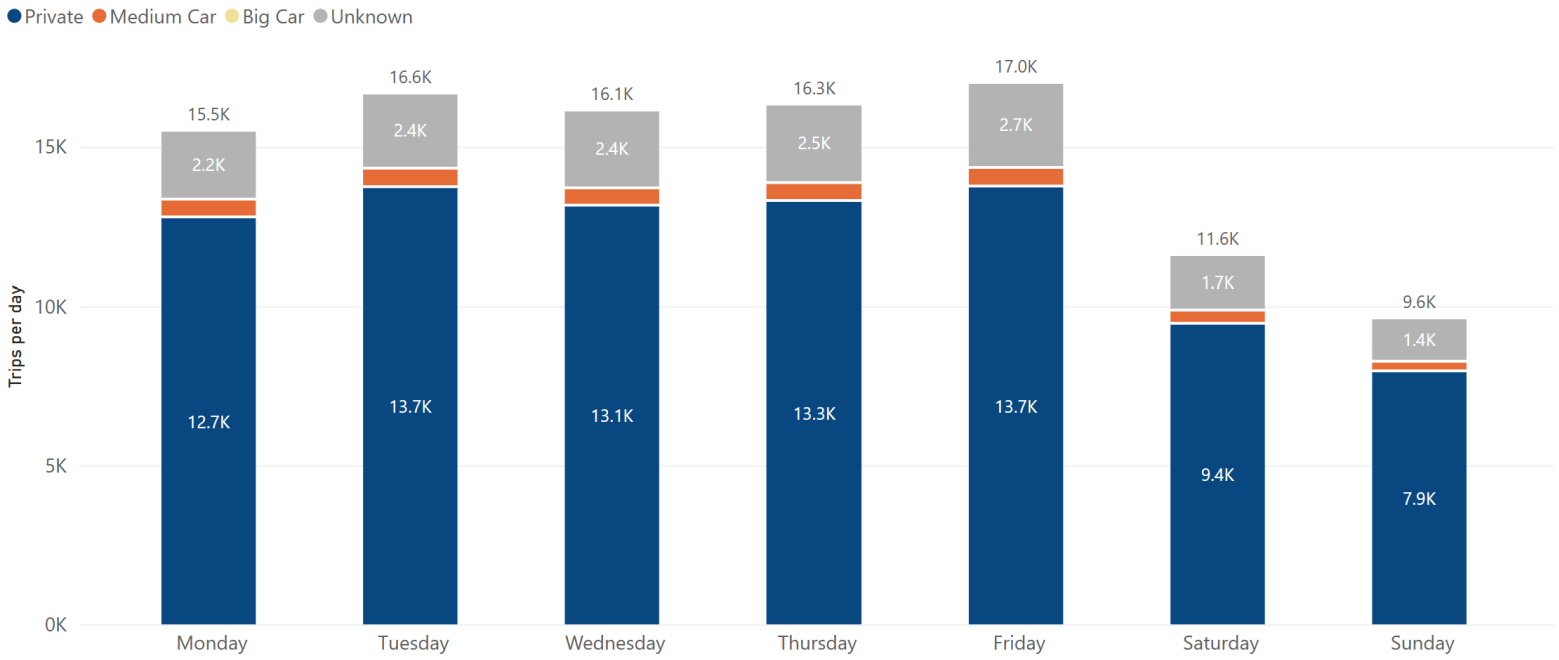
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Power BI report
891,523 trips · 4 pages

	Our data (trips per day)	TLC official
May 2020	11,153	11,237
June 2020	18,192	18,326

Our counts are slightly lower because of the 6,608 excluded trips. Every number in this deck comes from a SQL query and matches the Power BI report.

Private cars are about 82% of trips every day; Friday and Tuesday are the busiest



80.8–82.6%

Private, on every weekday

16,978

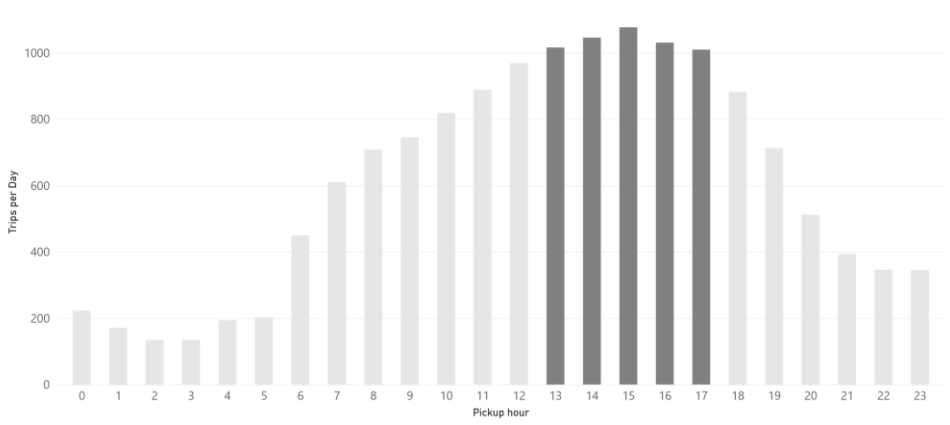
trips per day on Friday, the busiest

9,594

trips per day on Sunday, the quietest

The vehicle mix does not depend on the day; only the volume does.

Demand peaks 13:00–17:00 on weekdays, with no morning rush



weekday	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Monday	207	141	116	114	197	235	520	718	835	877	908	968	1018	1075	1107	1139	1111	1111	917	686	475	374	317	306
Tuesday	192	145	125	137	207	240	543	734	891	966	1019	1094	1187	1179	1189	1217	1191	1145	972	739	505	376	329	321
Wednesday	187	147	123	145	217	236	530	724	874	901	982	1037	1086	1200	1173	1182	1101	1114	942	721	490	361	313	323
Thursday	193	162	131	153	220	239	557	734	863	889	977	1023	1133	1147	1164	1206	1145	1079	950	748	519	384	337	345
Friday	214	173	130	144	213	226	544	728	869	872	931	991	1100	1143	1183	1230	1217	1242	1077	866	603	450	402	428
Saturday	289	221	163	141	169	137	265	375	387	450	565	657	718	774	850	857	772	727	692	658	508	423	379	384
Sunday	271	205	148	106	136	112	208	285	271	295	383	480	570	629	680	732	699	668	639	576	479	378	339	303

35.4%

of trips start between 13:00 and 17:00

1,077

trips per day at 15:00, the peak hour

4.2–5.1%

of trips in each hour from 07:00 to 09:00

Most cars are needed on weekday afternoons, not at the usual commute times.

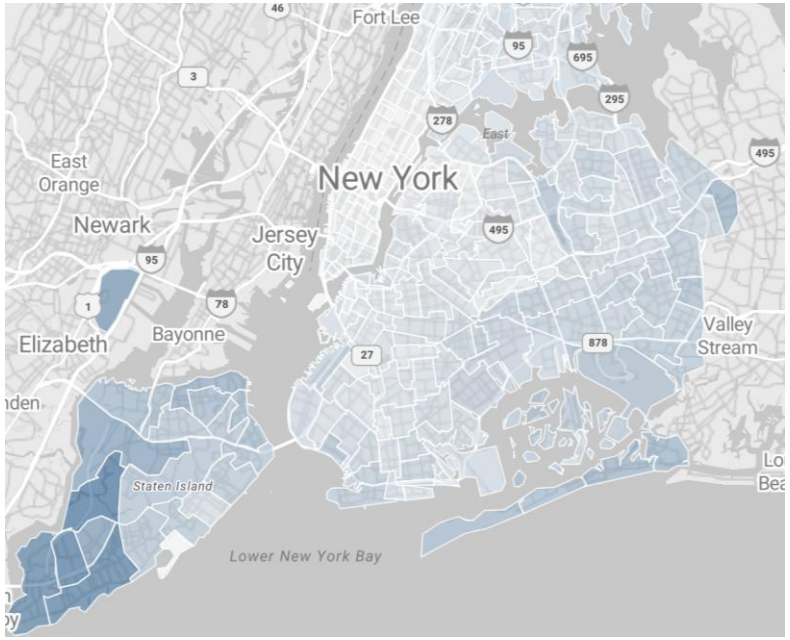
NYC Yellow Taxi · May–June 2020 · Source: NYC TLC trip records

Manhattan has 86% of pickups, but trips that start outside it pay about twice as much

Pickups



Avg driver income per trip



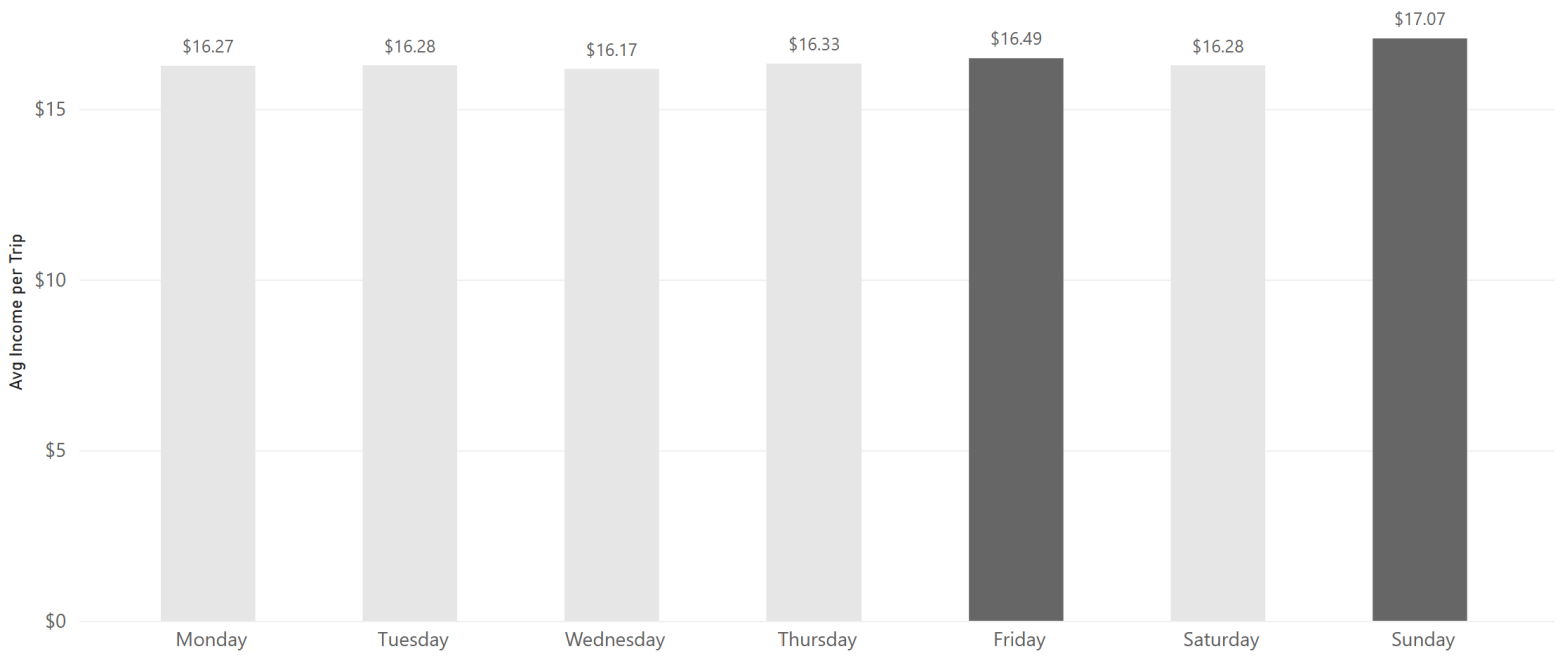
86.37%
of pickups are in Manhattan

$33.68
per trip from Queens, vs $14.17 from Manhattan

$48.20
per JFK pickup; airports are only 2.71% of pickups

Manhattan brings the volume; airports and outer boroughs bring the value per trip.

Sunday and Friday pay the most per trip, but the gap between days is only 5.5%



$17.07
average driver income per trip on Sunday

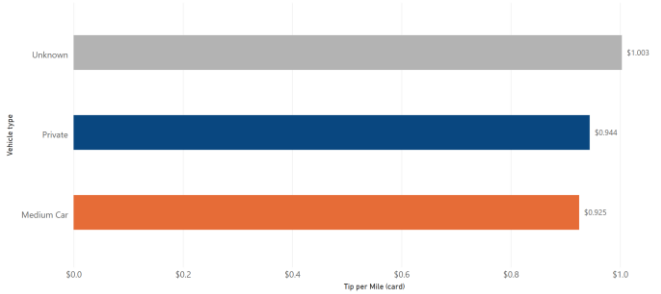
$16.49
on Friday

$0.90
gap between the best and the worst day (Wednesday)

For a driver, the day of the week matters much less than the hour or the place.

Tips per mile are about the same for every vehicle type once only card tips are counted

Vehicle type	All trips	Card only
Unknown	$0.168	**$1.003**
Private	$0.614	**$0.944**
Medium Car	$0.616	**$0.925**

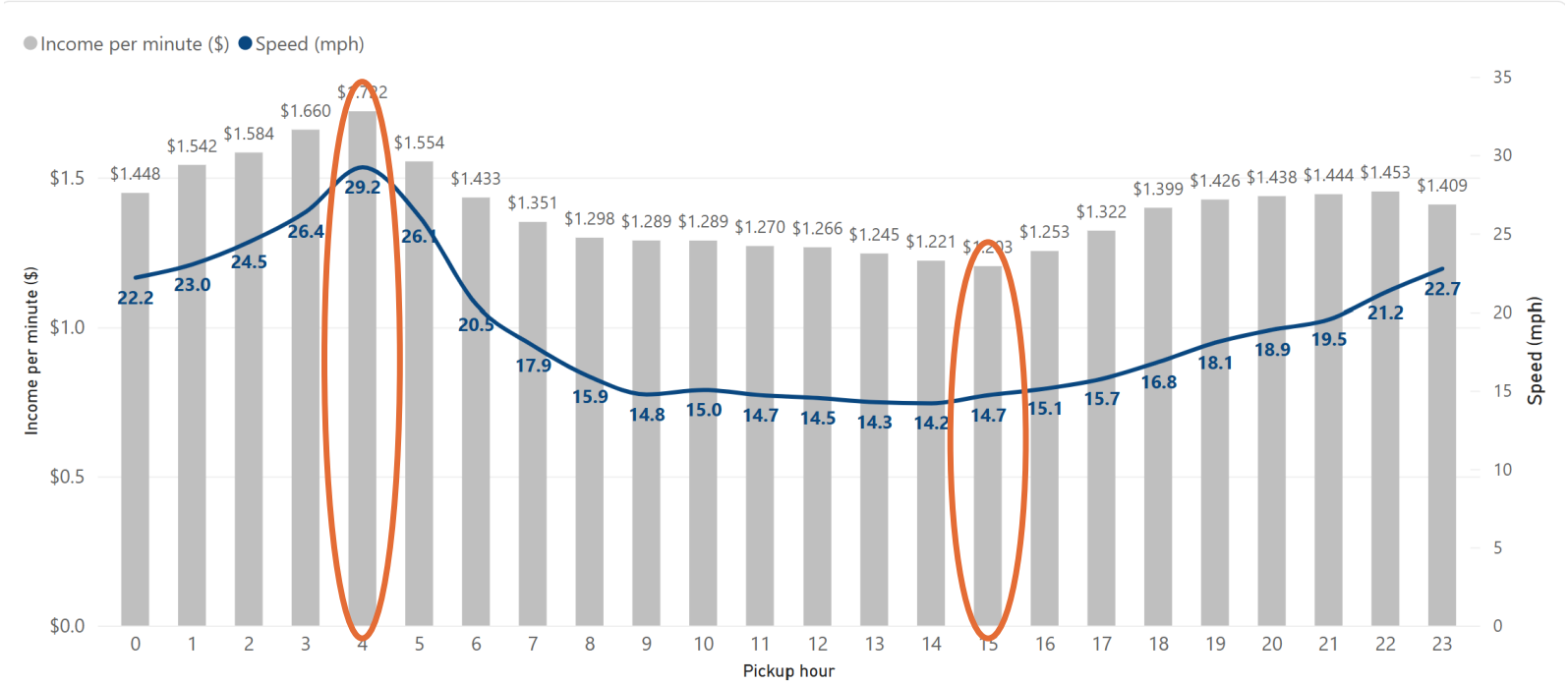


$0.945

tip per mile, all card payments

The first result came from the payment type, not from the vehicle type: cash tips are not recorded in TLC data.

Night hours pay the most per minute; slow afternoon traffic pays the least



$1.722

per minute at 04:00, the best hour

$1.203

per minute at 15:00, the worst hour

29 vs 14 mph

traffic speed at 04:00 vs 14:00

The busiest hours are the slowest, so they pay the least per minute ($1.248 vs $1.390 for other hours).

What to do with these results

Vendor

• Plan the most cars for weekday afternoons, 13:00–17:00 (35.4% of trips), and for Friday and Tuesday.
• Keep supply in Manhattan (86.37% of pickups); treat airports as few trips with high value ($48.20 per JFK pickup).
• Plan for empty returns: Brooklyn, the Bronx and Queens receive more trips than they send.

Driver

• Per minute, night hours pay the most ($1.722 at 04:00), but there are fewer passengers.
• The JFK flat fare pays $61.29 per trip for 27.81 minutes (6,693 trips).
• The weekday matters little (5.5% gap); the hour and the place matter more.

Limits of this analysis and next steps

Limits

- Cash tips are not recorded, so tip results use card payments only.
- Minutes are trip time only; waiting time between trips is not in the data.
- There is no vehicle or driver ID, so income per shift or per car is unknown.
- May–June 2020 was the start of the recovery after the lockdown; the results describe that period only.

Next steps

- Repeat the analysis on a full year of TLC data to compare with normal demand.
- Add the drop-off zone to the income analysis to measure empty returns.